

ARM-450 rev I

Assembly guide -- from the base to the tool flange

6-DOF arm, 6 x Waveshare ST3215 servos on one bus.
27 printed files (33 pieces), 199 bought items, 114 screws.

HOW THIS GUIDE WORKS

24 steps, from the foot up to the tool flange. Each page shows the arm so far in GREY and what the step adds in colour (red printed part, black servo, blue screws / bearings / inserts) -- whole (left) and close up (right) -- then what you need and do. BENCH steps put one sub-assembly together on the table right before it goes on the arm.

The order is PROVEN: a hex key reaches every screw at the step where it is tightened (checked on the exact 3D model):

114 screws -- 0 blocked at their step

(B2F_ORDER_CHECK.log -- LOGS/ in the release, REV_H/ in the source)

BEFORE STEP 1 -- pages 2 to 5

- * print the fit test; coupon 07 must screw onto a servo's back
- * heat-set inserts and bearings into the printed parts
- * give every servo its own ID and centre it
- * cut and crimp the 6 bus cables to length

SAFETY

Until the springs are on (step 24) the links swing freely when the servos are unpowered: support the arm by hand.
Screw the foot to the table in step 1 -- the arm tips at reach.

Tools and bought parts

TOOLS

- * 1.5 mm hex key
- * 2.0 mm hex key
- * 2.5 mm hex key
- * 3.0 mm hex key
- * 5.5 mm spanner
- * Phillips PH1 screwdriver
- * vise or clamp + retaining compound (e.g. Loctite 641)
- * soldering iron with an insert tip (heat-set inserts)
- * long-nose pliers (bus plugs at J1, J4, J6)
- * the servo tool (Waveshare) to set IDs and centre the servos

BOUGHT PARTS

6706 bearing (30x37x4)	4
6806 bearing (30x42x7)	6
M2.5 heat-set insert	2
M2.5 x 16 socket head	2
M2.5 x 6 socket head (horn)	24
M3 heat-set insert	55
M3 nyloc nut	4
M3 x 10 socket head	12
M3 x 12 socket head	16
M3 x 16 socket head	10
M3 x 20 socket head	1
M3 x 4 set screw	1
M3 x 45 socket head	16
M4 heat-set insert	4
M4 x 12 socket head	4
M5 x 10 set screw (cup point)	8
ST3215 servo	6
extension spring J2 (7.4 N @ 36.0 / 12.4 N @ 58.2 mm, inside hooks)	2
extension spring J3 (2.2 N @ 43.0 / 5.0 N @ 91.4 mm, inside hooks)	2
self-tapping screw ~2.2 x 8 pan head (from the servo box)	20

Preparation 1 -- print, fit test first

Print 00_FIT_TEST_print_first first (7 coupons, about 2 h) with the settings you will use for the arm. Coupon 07 is a copy of a servo cover's back: lay it on the back of a servo and drive the 4 self-tapping screws from the servo box -- they must line up, bite and pull it flat. If a bearing does not press into coupons 02/03, the shaft stub 04 does not slide through a bearing, or the servo rattles in coupon 01, tune the slicer's XY compensation before the big parts.

file	qty	orient	supports
01_base_x1.stl	1	+Z up	REQUIRED
02_spigot_collar_x1.stl	1	+Z up	none
03_j1_mount_x1.stl	1	+Z up	none
04_j1_hub_x1.stl	1	+Z up	advised
05_J2_turret_p1_x1.stl	1	+X up	REQUIRED
06_J2_turret_p2_x1.stl	1	-Y up	advised
07_J2_shaft_x1.stl	1	+Y up	none
08_shaft_clamp_x4.stl	4	-Z up	none
09_link_upper_groove_x1.stl	1	+Z up	advised
10_link_upper_tongue_x1.stl	1	+Z up	advised
11_collar_upper_x2.stl	2	+Z up	none
12_J3_p1_x1.stl	1	-Z up	REQUIRED
13_J3_p2_x1.stl	1	-Z up	none
14_J3_shaft_x1.stl	1	-Z up	none
15_link_fore_tongue_x1.stl	1	+Z up	advised
16_link_fore_groove_x1.stl	1	+Z up	advised
17_collar_fore_x2.stl	2	+Z up	none
18_j4_base_x1.stl	1	-Z up	REQUIRED
19_j4_cap_x1.stl	1	-Z up	none
20_j4_hub_x1.stl	1	-Z up	none
21_J5_p1_x1.stl	1	+X up	advised
22_J5_p2_x1.stl	1	-Z up	none
23_J5_shaft_x1.stl	1	-Z up	none
24_J5_spacer_x2.stl	2	+Z up	none
25_j6_body_x1.stl	1	-X up	REQUIRED
26_j6_cap_x1.stl	1	-X up	none
27_j6_flange_x1.stl	1	-X up	none

Preparation 2 -- inserts and bearings

HEAT-SET INSERTS (flush with the surface, square):

part	heat-set inserts
J2_turret_p1	6 x M3
J3_p1	6 x M3
J5_p1	4 x M3
collar_fore_+y	1 x M3
collar_fore_-y	1 x M3
collar_upper_+y	1 x M3
collar_upper_-y	1 x M3
j1_mount	4 x M4
j4_base	4 x M3
j4_hub	4 x M3
j6_body	4 x M3
j6_flange	4 x M3
link_fore_groove	1 x M2.5, 8 x M3
link_fore_tongue	2 x M3
link_upper_groove	1 x M2.5, 6 x M3
link_upper_tongue	2 x M3
spigot_collar	1 x M3

BEARINGS (press on the OUTER ring; a drop of retaining compound on it):

base	2 x 6806	(J1)
J2_turret_p1	2 x 6806	(J2, from the fork gap)
J3_p1	2 x 6806	(J3, from the fork gap)
j4_cap	1 x 6706	(J4, from below)
J5_p1	2 x 6706	(J5, from the fork gap)
j6_cap	1 x 6706	(J6, from below)

Preparation 3 -- servos and cables

SERVO IDs: all six servos share one bus, so each needs its own ID. Connect ONE servo at a time to the Waveshare servo tool and give it ID 1 (J1) to 6 (J6); label it. New servos usually all carry the same ID.

CENTRE: before a horn coupler (hub, shaft, axle, flange) is bolted on, move that servo to its centre, position 2048 of 4096, and bolt the coupler with its link at home (straight up). Then the software's zero is the arm's home and the +- ranges on the last page are symmetric.

CABLES: one bus cable per joint (daisy chain controller -> J1 -> ... -> J6), 3-pin bus cables. Lengths and the loop to leave at each joint (measured on the model):

cable	crosses	range	loop to leave	cable length
J1 -> J2	J1	+90.0	54 mm	>= 263 mm
J2 -> J3	J2	+54.0	37 mm	>= 267 mm
J3 -> J4	J3	+72.0	33 mm	>= 219 mm
J4 -> J5	J4	+90.0	62 mm	>= 240 mm
J5 -> J6	J5	+46.5	31 mm	>= 146 mm

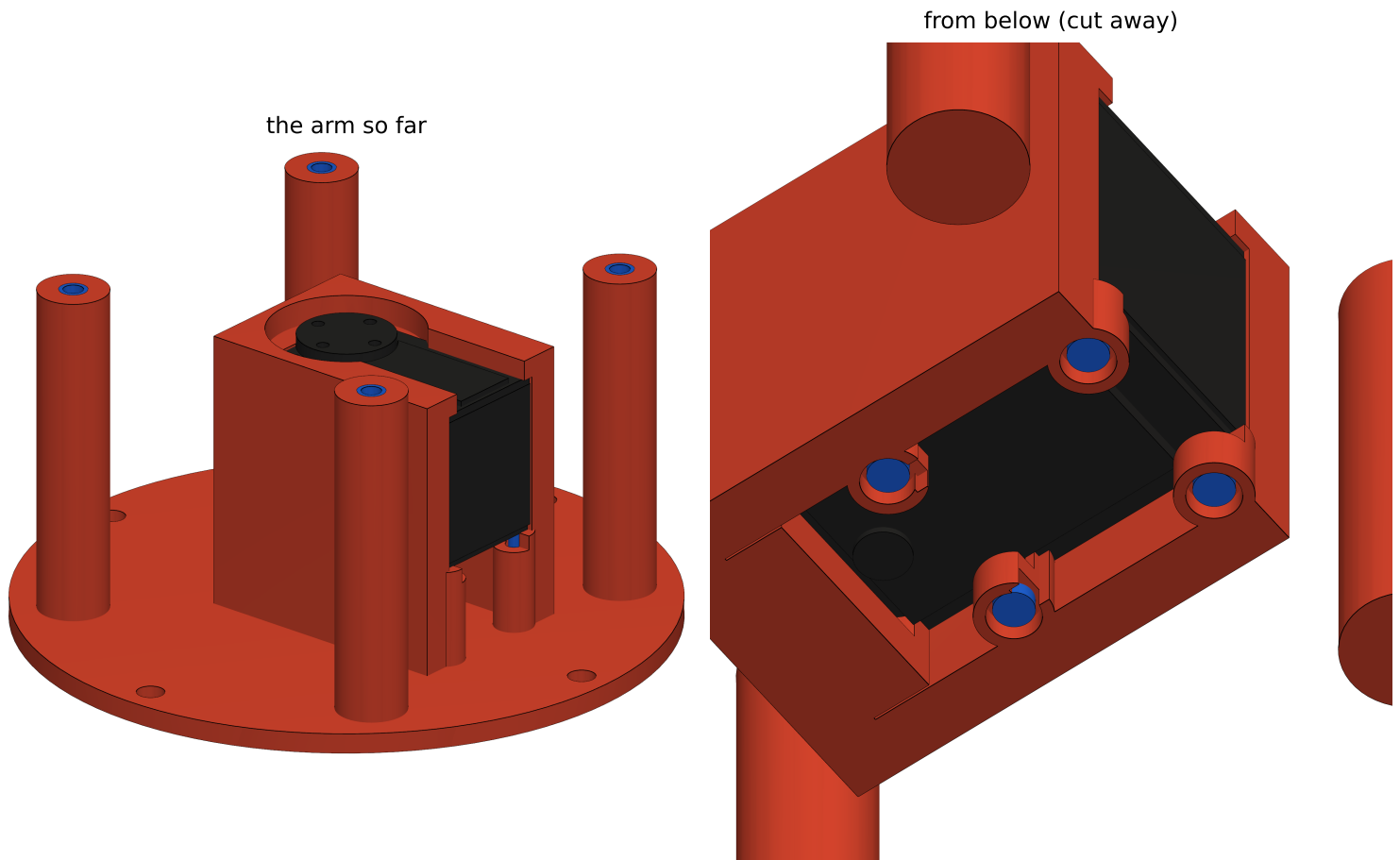
controller -> J1: both in the foot, no joint crossed

J6: last on the bus, no cable leaves it

Plug slots: every plug can go in and out with the arm assembled (J2/J3/J5 straight through the cover window; J1/J4/J6 lift 10-14 mm, then sideways -- use pliers). Easiest is still to plug in before a servo goes in.

Step 1 of 24 (ON THE ARM)

Foot: J1 servo into the foot, its 4 screws from under the foot plate



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: j1_mount
4 x M4 heat-set insert (set in preparation 2)
1 x ST3215 servo
4 x self-tapping screw ~2.2 x 8 pan head (from the servo box)
TOOLS: Phillips PH1 screwdriver

DO

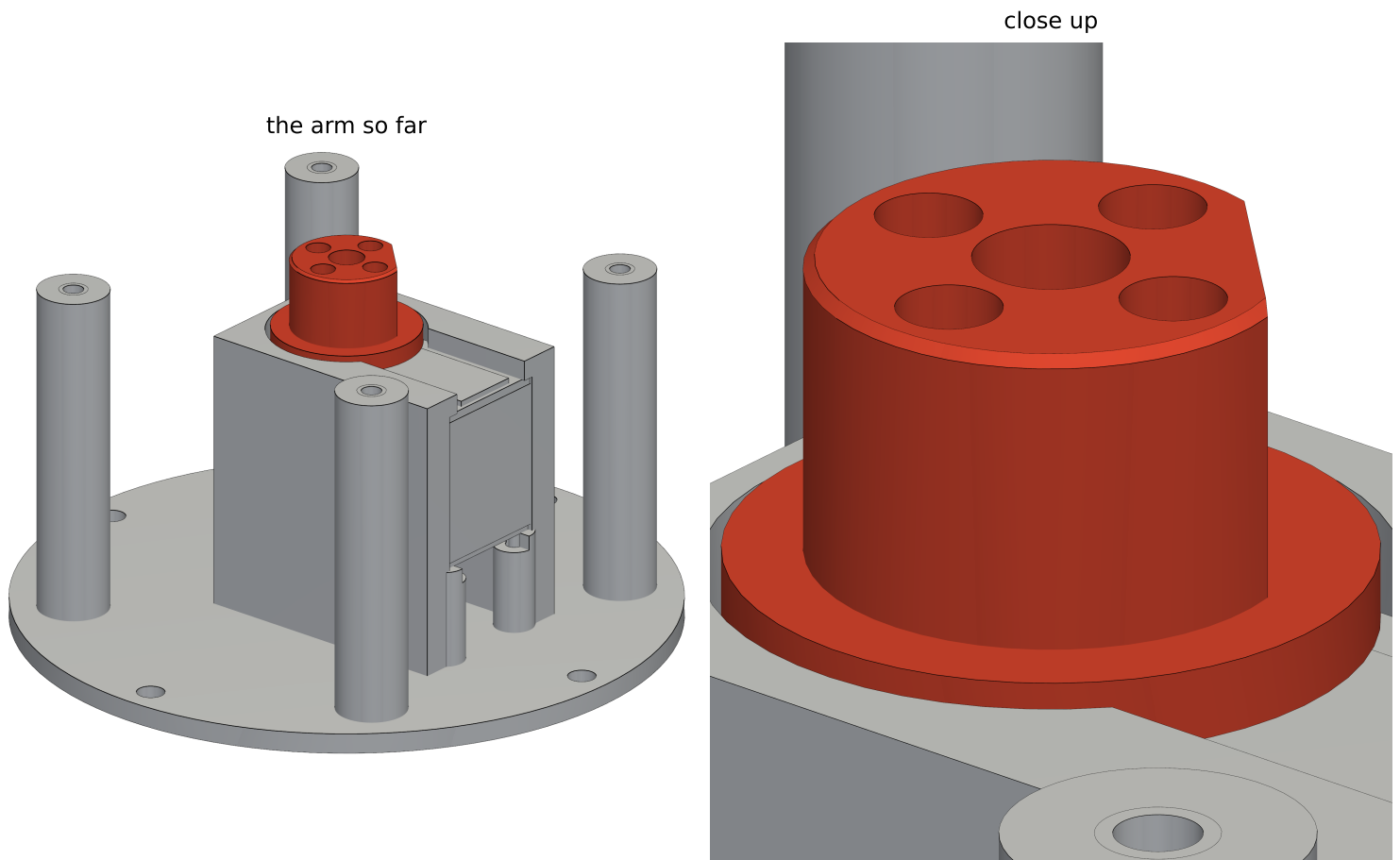
Plug the controller's bus cable into one of the J1 servo's two sockets (the other takes the cable to J2 later; both can also be plugged in afterwards -- see the plug slots). Slide the J1 servo into the foot's channel from the open end, horn UP, until its end stops on the end wall; its front face slides under the two lips. Turn the foot over and drive the 4 self-tapping screws from the servo's box up through the 4 post holes into the servo's back holes: snug, do not strip -- 2.6 mm go into the servo. THEN screw the foot to the table: 4 x M5 bolts or #10 wood screws through the Ø5.5 holes (the base covers them in step 3).

CHECK

The servo stands on its 4 posts and does not move when pushed; the foot does not rock on the table.

Step 2 of 24 (ON THE ARM)

J1 hub onto the J1 servo horn



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: j1_hub
4 x M2.5 x 6 socket head (horn)
TOOLS: 2.0 mm hex key

DO

Centre the J1 servo (position 2048 of 4096) with the servo tool. Put the J1 hub on the horn so its D-flat sits between two horn holes, as in the close-up (45 deg -- it keys the turret's heading); 4 x M2.5 x 6 through the hub flange into the horn.

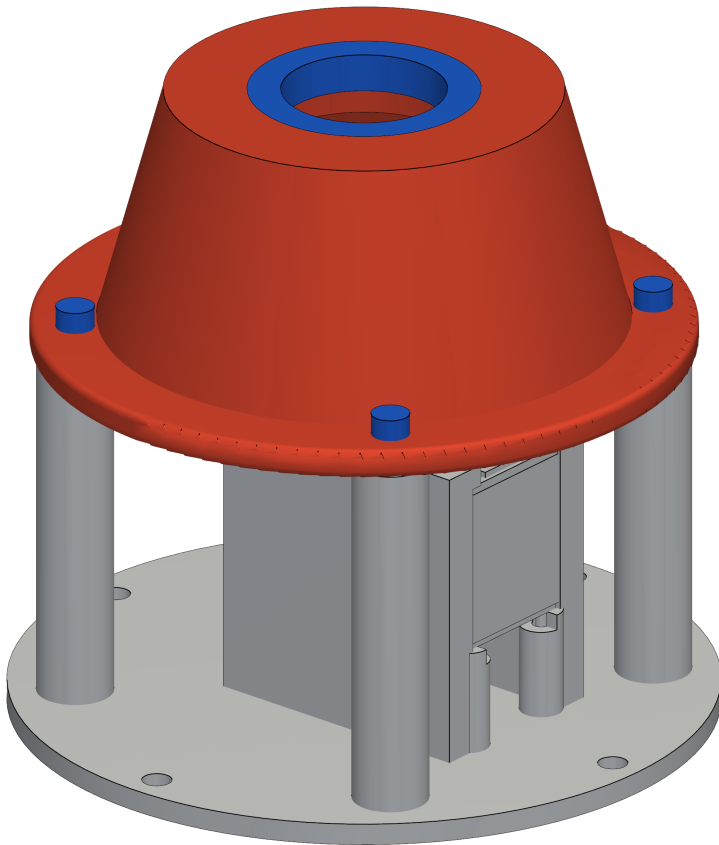
CHECK

The hub flange turns inside the lip opening without touching.

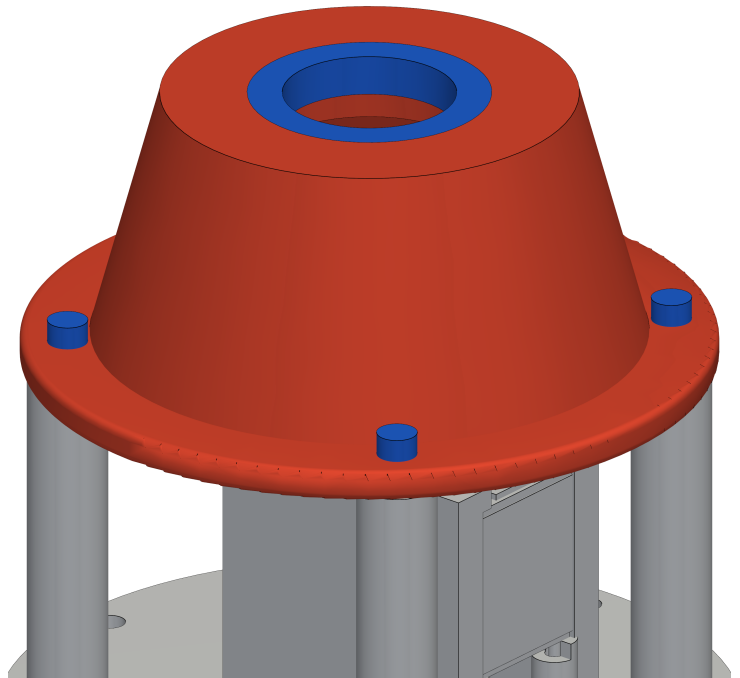
Step 3 of 24 (ON THE ARM)

Base with the two J1 bearings onto the foot

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: base

2 x 6806 bearing (30x42x7) (pressed in: preparation 2)

4 x M4 x 12 socket head

TOOLS: 3.0 mm hex key, vise or clamp + retaining compound (e.g. Loctite 641)

DO

Press the two 6806 bearings into the base, the lower one from underneath (a drop of retaining compound on each outer ring; press on the OUTER ring only). Set the base on the foot's four columns and bolt it: 4 x M4 x 12 into the column inserts.

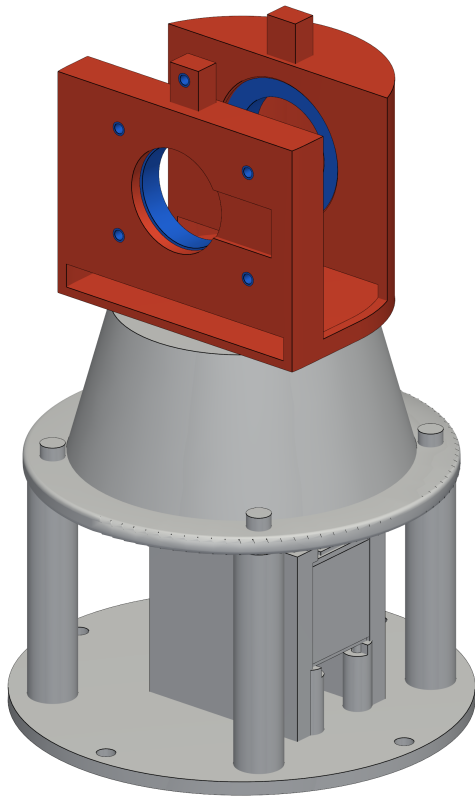
CHECK

Both bearings' inner rings turn freely by finger.

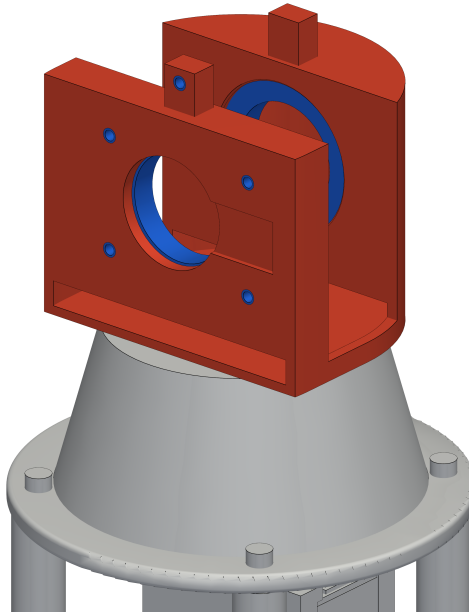
Step 4 of 24 (ON THE ARM)

Turret onto the J1 hub, spigot collar

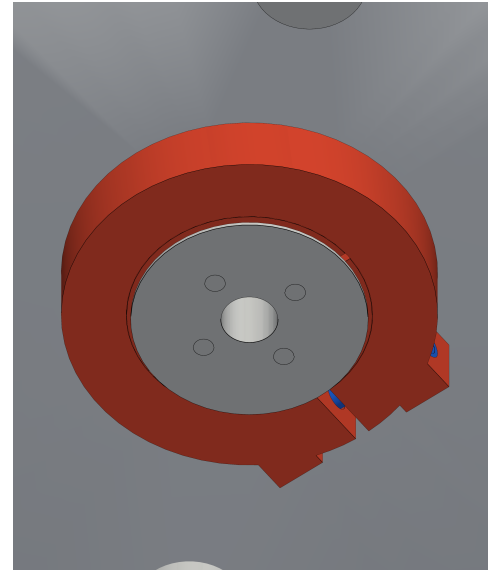
the arm so far



close up



from below (cut away)



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J2_turret_p1, spigot_collar
2 x 6806 bearing (30x42x7) (pressed in: preparation 2)
7 x M3 heat-set insert (set in preparation 2)
1 x M3 x 20 socket head
TOOLS: 2.5 mm hex key, vise or clamp + retaining compound (e.g. Loctite 641)

DO

Turret p1 has its two 6806 (J2) pressed in and its inserts set (preparation). Lower the turret's spigot through the base bearings onto the J1 hub: turn it until the spigot's D-key drops over the hub's D-flat (turret facing forward). From below the base, slide the spigot collar over the slit end of the spigot and fit its M3 x 20 pinch bolt into the collar insert. Tighten the pinch bolt LAST: it closes the collet onto the hub plug.

CHECK

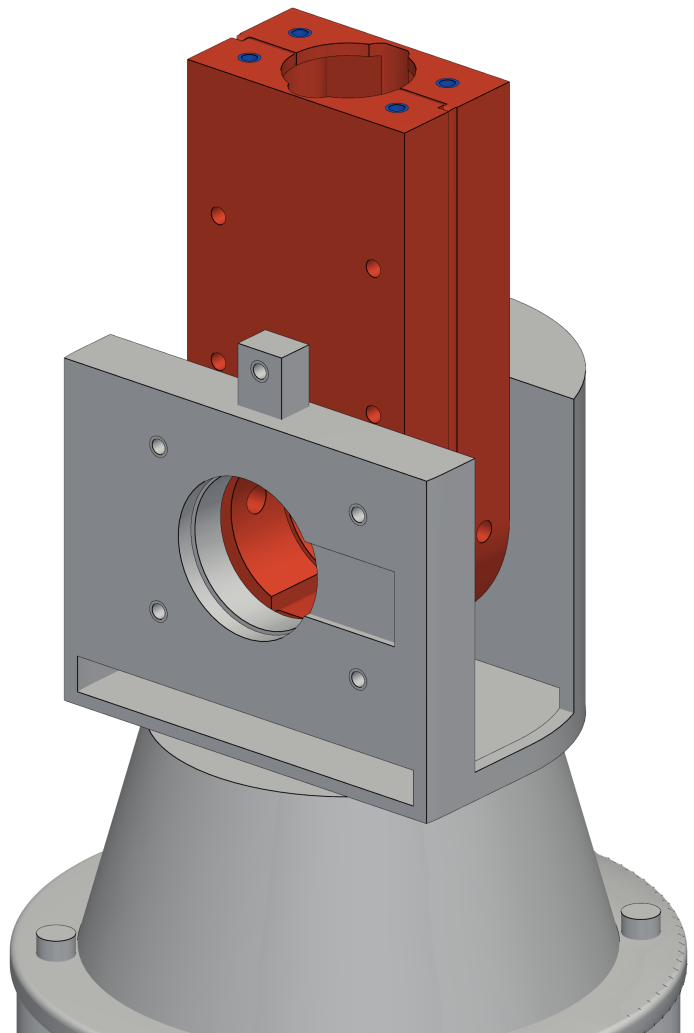
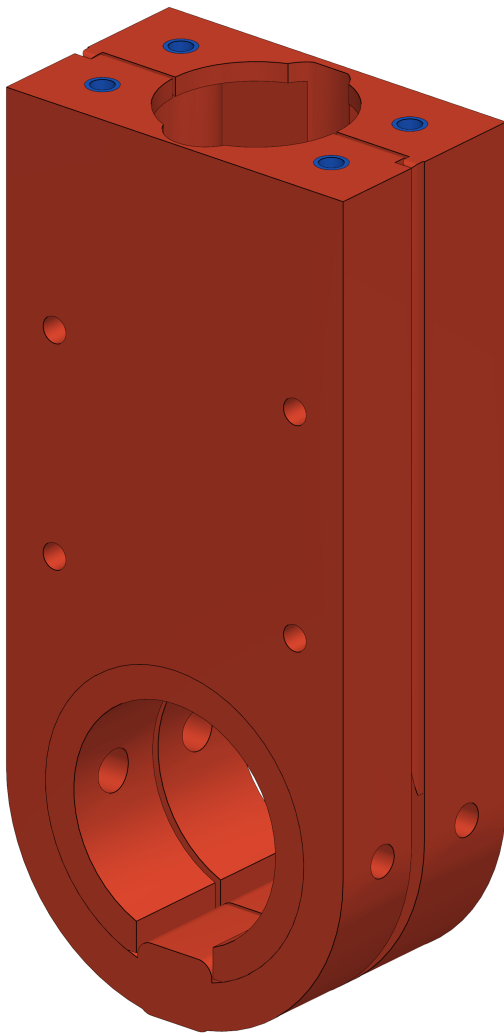
The turret turns with the J1 horn only -- no rotational play, no lift.

Step 5 of 24 (ON THE BENCH)

upper link halves round their centring rings

on the bench

where it goes on the arm



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: link_upper_groove, link_upper_tongue, shaft_clamp_1, shaft_clamp_2

1 x M2.5 heat-set insert (set in preparation 2)

1 x M2.5 x 16 socket head

8 x M3 heat-set insert (set in preparation 2)

4 x M3 x 16 socket head

TOOLS: 2.0 mm hex key, 2.5 mm hex key

DO

Centring rings (shaft_clamp) into both upper-link halves, the ring's gap over the seam

ear. Run an M5 screw once into each 04.2 side hole to cut its thread, and out again.

Close the halves: 4 x M3 x 16 seam screws (heads in the tongue's counterbores) into the groove half's inserts, and the M2.5 x 16 ear bolt.

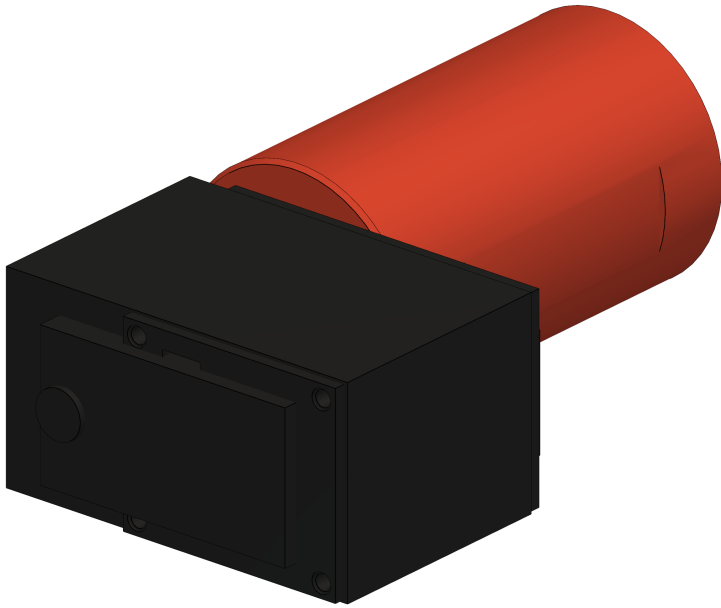
CHECK

The halves are flush all round; the Ø30 bore is round and the rings do not move.

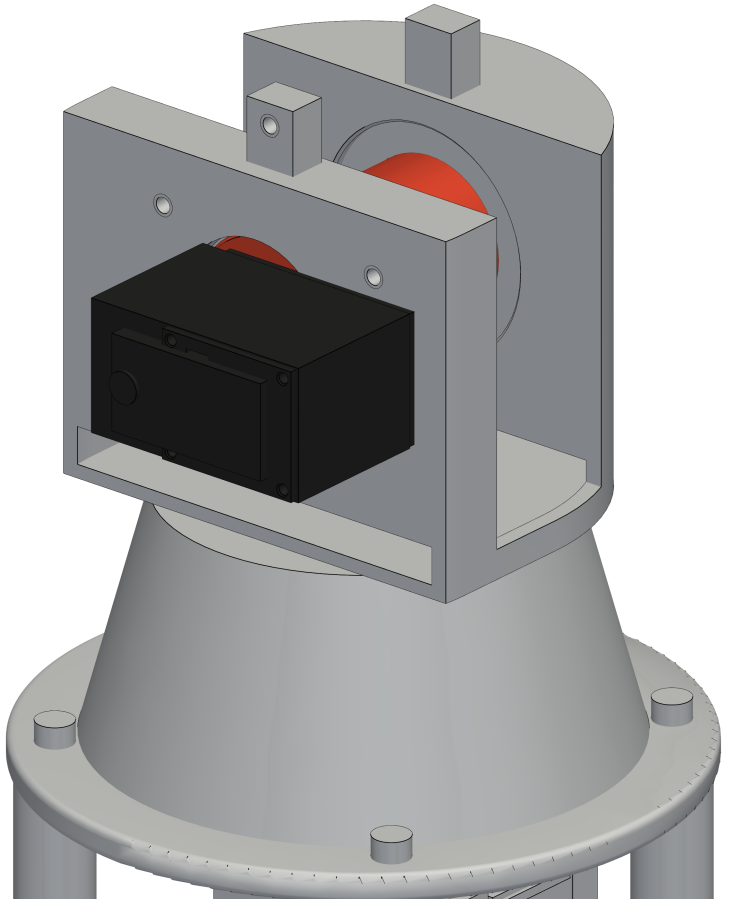
Step 6 of 24 (ON THE BENCH)

J2 servo onto its printed shaft

on the bench



where it goes on the arm



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J2_shaft
4 x M2.5 x 6 socket head (horn)
1 x ST3215 servo
TOOLS: 2.0 mm hex key

DO

Centre the J2 servo (2048). Put the printed J2 shaft's closed end on the horn and drive 4 x M2.5 x 6 through the shaft's bore into the horn (a long 2.0 mm key). Plug the J1 -> J2 cable and the J2 -> J3 cable into the servo now (or later through the pod's plug slot).

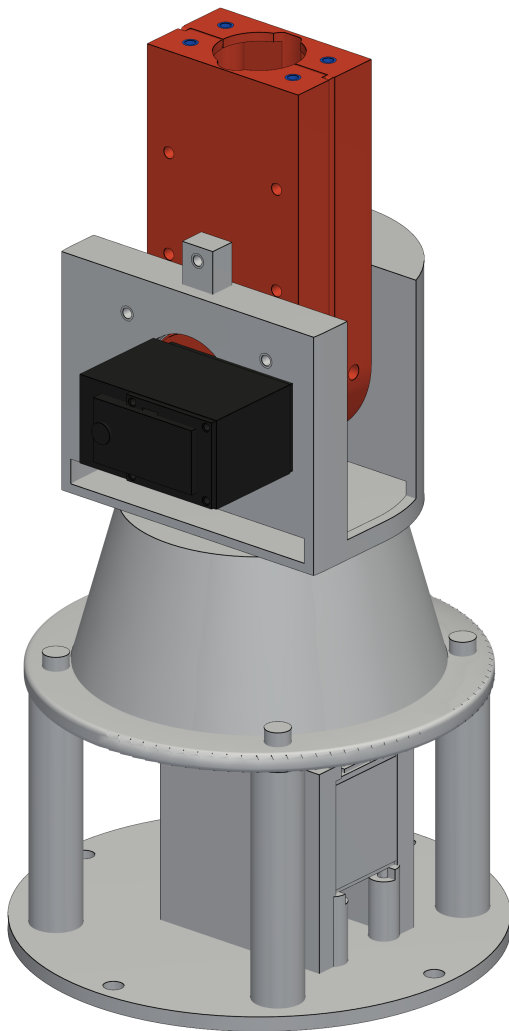
CHECK

The shaft runs true when the horn turns.

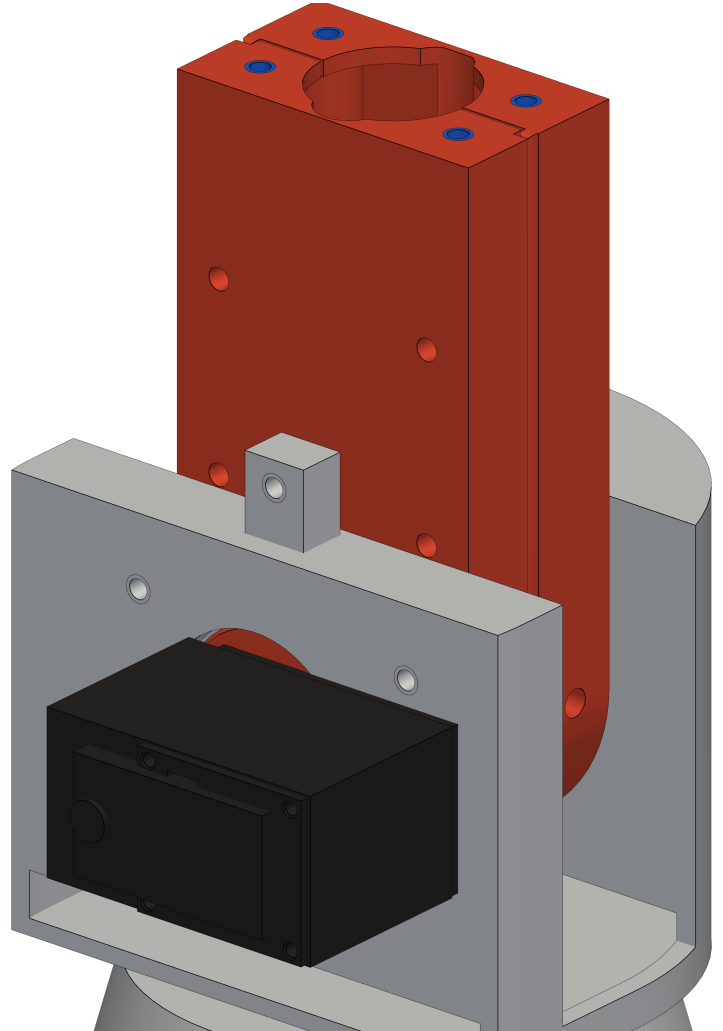
Step 7 of 24 (JOIN)

J2: upper link into the turret, servo + shaft down the axis

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED
(nothing new)

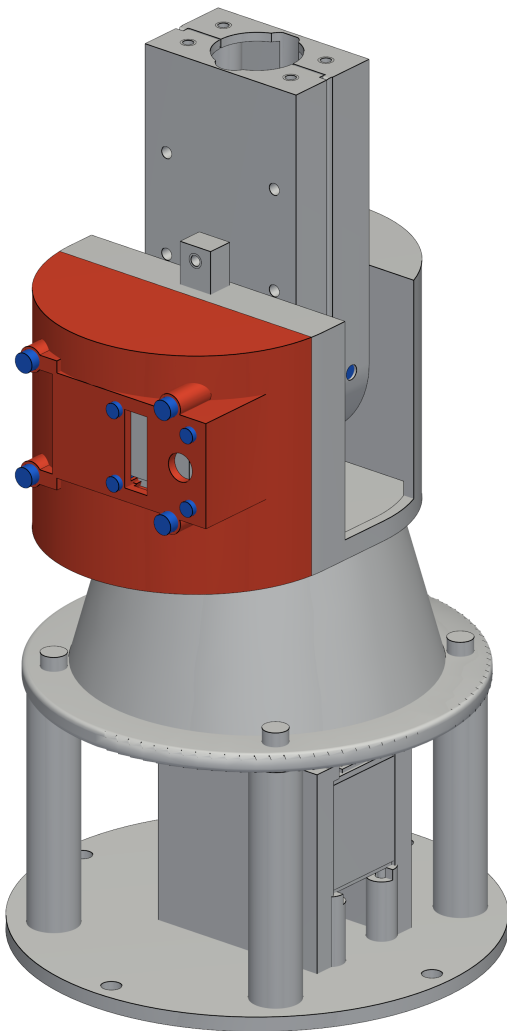
DO
Hold the upper link in the turret's fork gap, straight up (home), its bore on the J2 axis. Push the J2 servo + shaft down the axis from the pod side: the shaft goes through the drive bearing, the link's two centring rings and the idle bearing; the servo slides fully into its bay (a firm push: the bay pinches it).

CHECK
The servo is fully down in its bay; the link swings freely on the shaft (not locked yet).

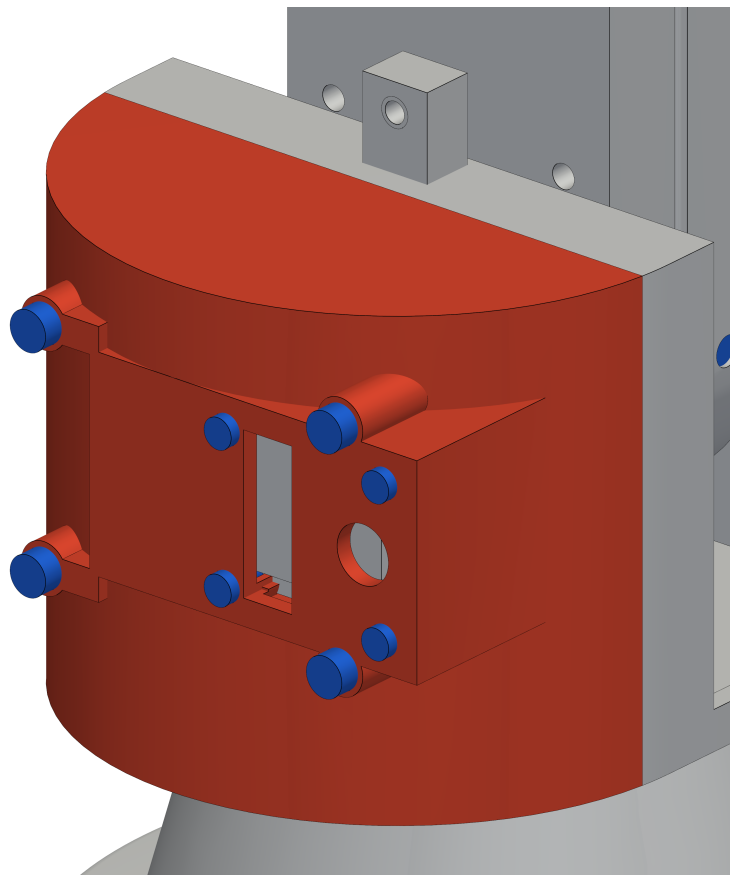
Step 8 of 24 (ON THE ARM)

J2: servo pod closed, servo screws, link set screws

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J2_turret_p2
4 x M3 x 45 socket head
4 x M5 x 10 set screw (cup point)
4 x self-tapping screw ~2.2 x 8 pan head (from the servo box)
TOOLS: 2.5 mm hex key, Phillips PH1 screwdriver

DO

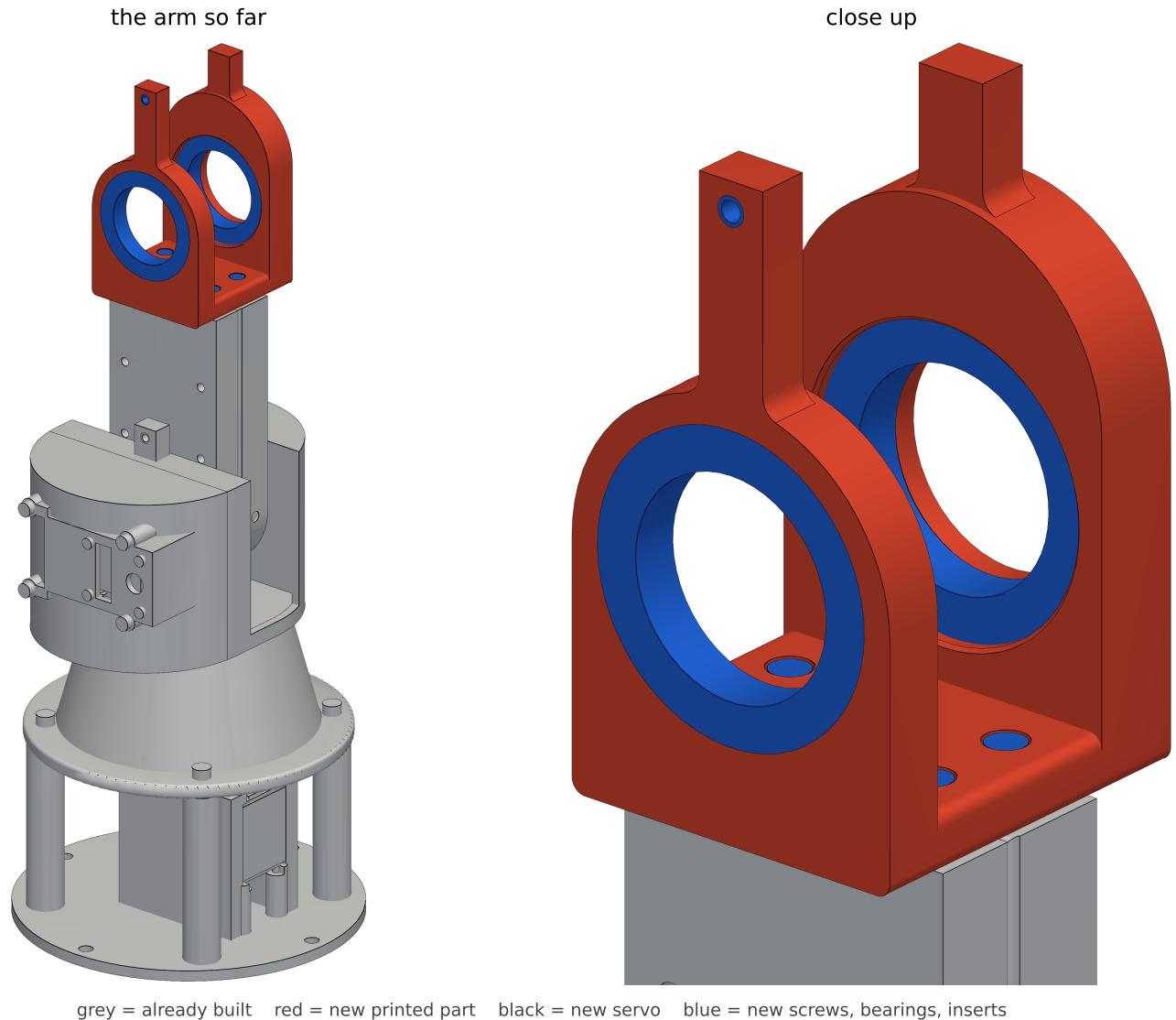
Lead both cables out through the pod's window. Close the pod (J2 turret p2): 4 x M3 x 45 into turret p1's inserts. Then 4 self-tapping screws from the servo box through the pod's back into the servo. With the servo centred and the link straight up, lock the link: 2 x M5 x 10 cup-point set screws per link half through the side holes onto the shaft's two flats. SUPPORT THE LINK from now on until the springs are on (step 24).

CHECK

The link moves only with the servo: no play when you rock it.

Step 9 of 24 (ON THE ARM)

J3 fork onto the top of the upper link



YOU NEED

printed: J3_p1

2 x 6806 bearing (30x42x7) (pressed in: preparation 2)

6 x M3 heat-set insert (set in preparation 2)

4 x M3 x 10 socket head

TOOLS: 2.5 mm hex key, vise or clamp + retaining compound (e.g. Loctite 641)

DO

J3 p1 has its two 6806 pressed in (preparation). Set it on the upper link's end face and bolt it: 4 x M3 x 10 into the link's end inserts -- the heads sit in the counterbores of the fork floor (they are under the swinging forearm).

CHECK

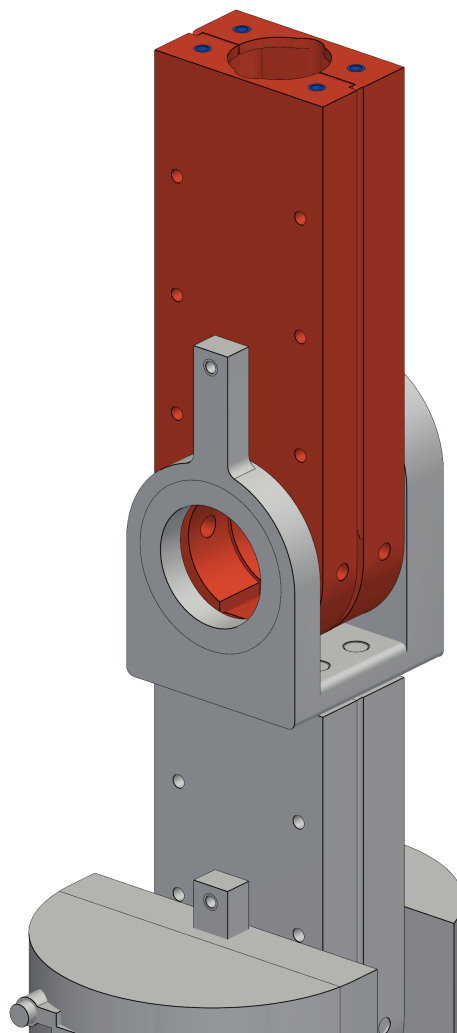
The heads are below the floor surface; the fork sits flat on the link.

Step 10 of 24 (ON THE BENCH)

forearm halves round their centring rings

on the bench

where it goes on the arm



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: link_fore_groove, link_fore_tongue, shaft_clamp_3, shaft_clamp_4

1 x M2.5 heat-set insert (set in preparation 2)

1 x M2.5 x 16 socket head

10 x M3 heat-set insert (set in preparation 2)

6 x M3 x 16 socket head

TOOLS: 2.0 mm hex key, 2.5 mm hex key

DO

Forearm halves round their centring rings, as the upper link in step 5: M5 thread once into each side hole, 6 x M3 x 16 seam screws, the M2.5 x 16 ear bolt.

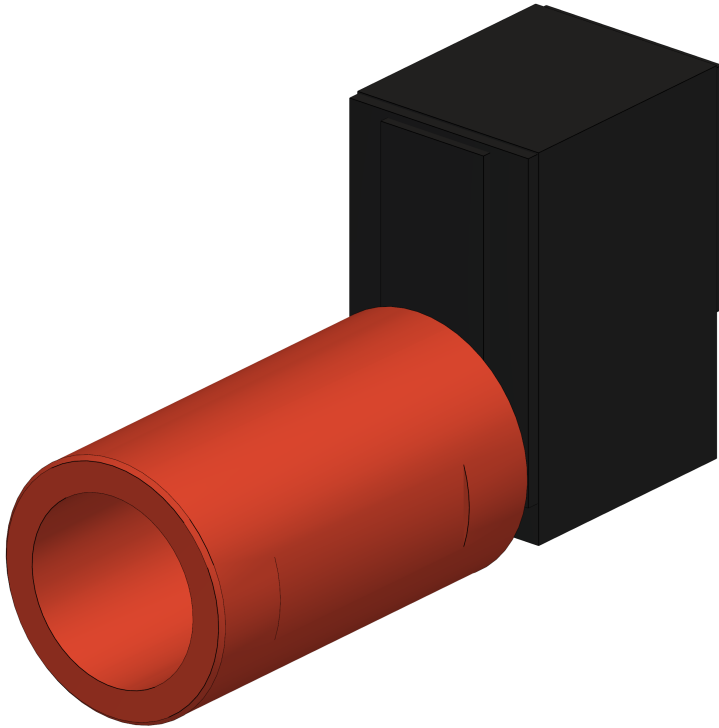
CHECK

Halves flush, bore round, rings fixed.

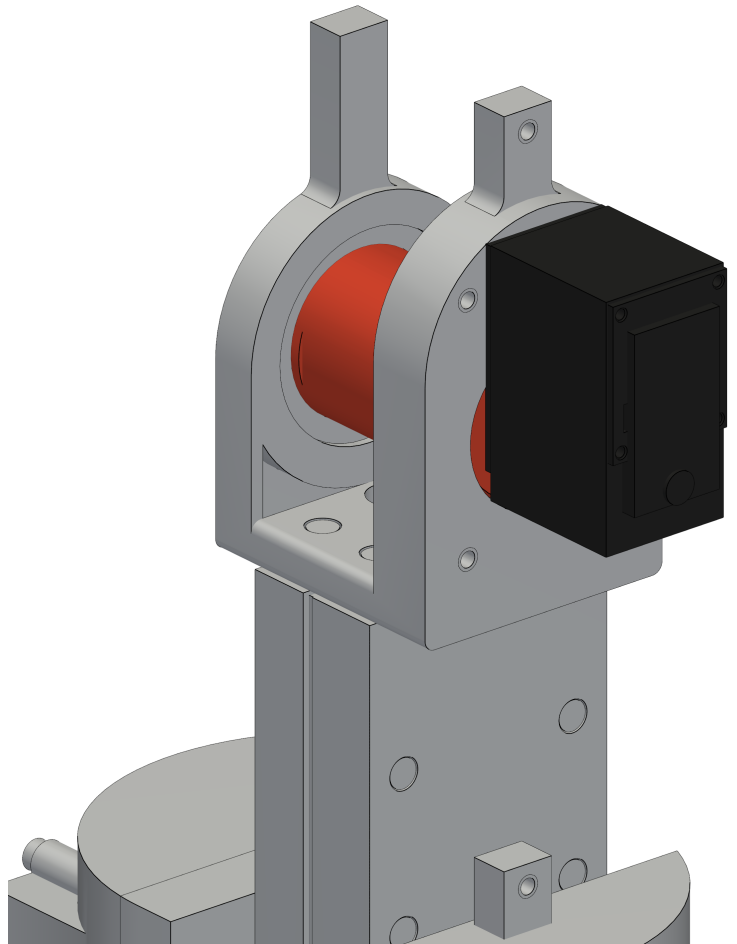
Step 11 of 24 (ON THE BENCH)

J3 servo onto its printed shaft

on the bench



where it goes on the arm



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J3_shaft
4 x M2.5 x 6 socket head (horn)
1 x ST3215 servo
TOOLS: 2.0 mm hex key

DO

Centre the J3 servo (2048); J3 shaft onto its horn with 4 x M2.5 x 6. Plug the J2 -> J3 and the J3 -> J4 cables in (or later through the cover's plug slot).

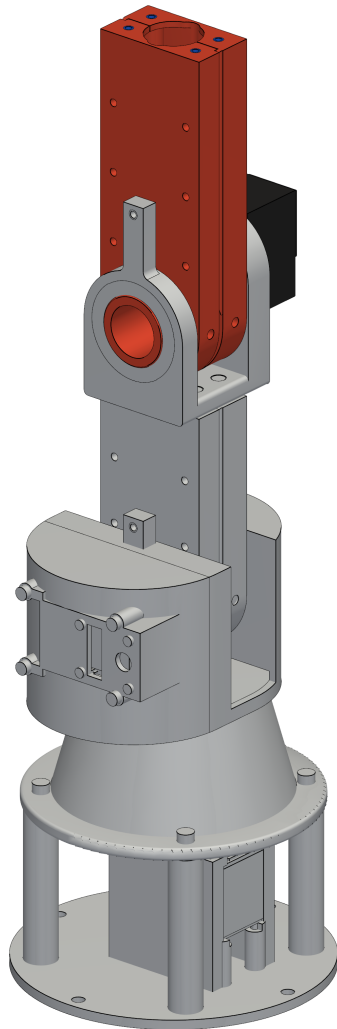
CHECK

The shaft runs true.

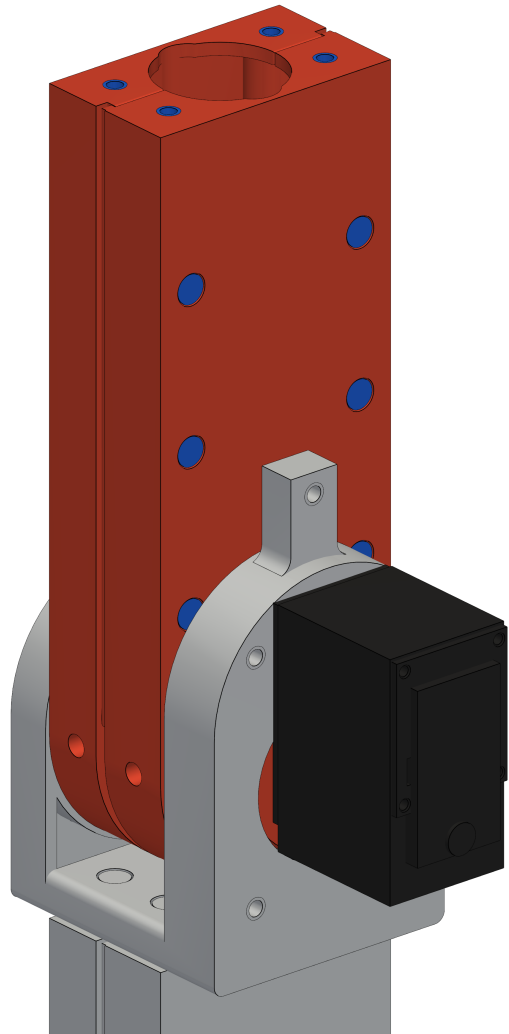
Step 12 of 24 (JOIN)

J3: forearm into the fork, servo + shaft down the axis

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED
(nothing new)

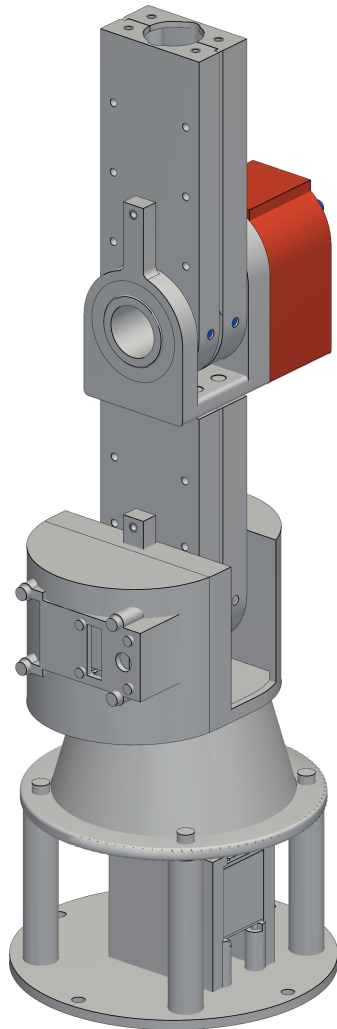
DO
Forearm into the J3 fork gap, straight (home), bore on the J3 axis; push the J3 servo + shaft down the axis through bearing, rings, bearing into the bay.

CHECK
Servo fully down; forearm swings freely (not locked yet).

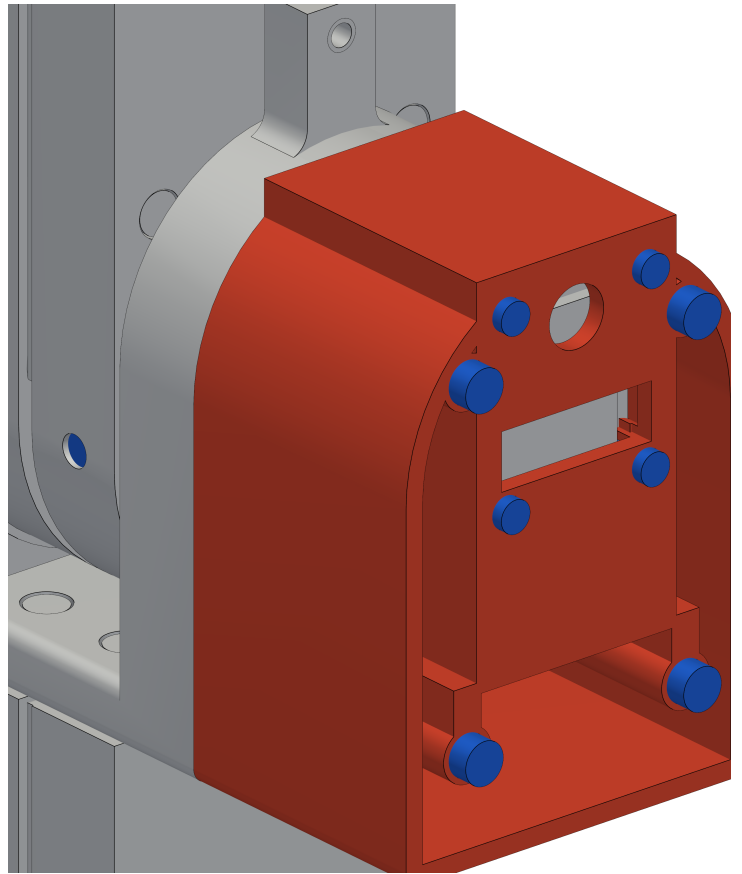
Step 13 of 24 (ON THE ARM)

J3: cover closed, servo screws, forearm set screws

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J3_p2

4 x M3 x 45 socket head

4 x M5 x 10 set screw (cup point)

4 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

TOOLS: 2.5 mm hex key, Phillips PH1 screwdriver

DO

Cables out through the cover's window; close the J3 cover: 4 x M3 x 45; 4 self-tapping servo screws through the cover; with the servo centred and the forearm straight, 2 x M5 x 10 set screws per forearm half onto the shaft's flats. Support the forearm until the springs are on.

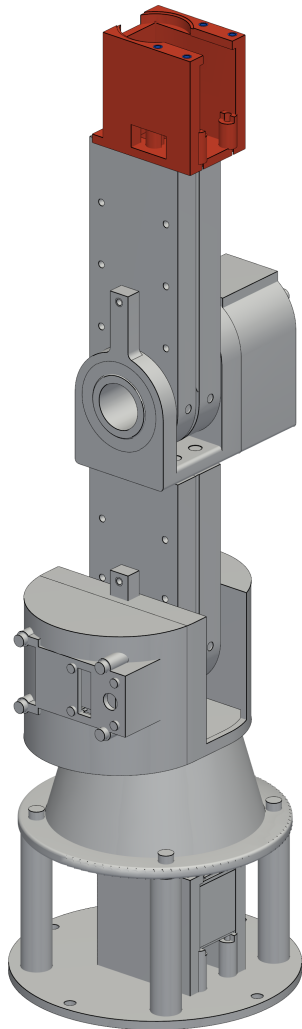
CHECK

The forearm moves only with the J3 servo.

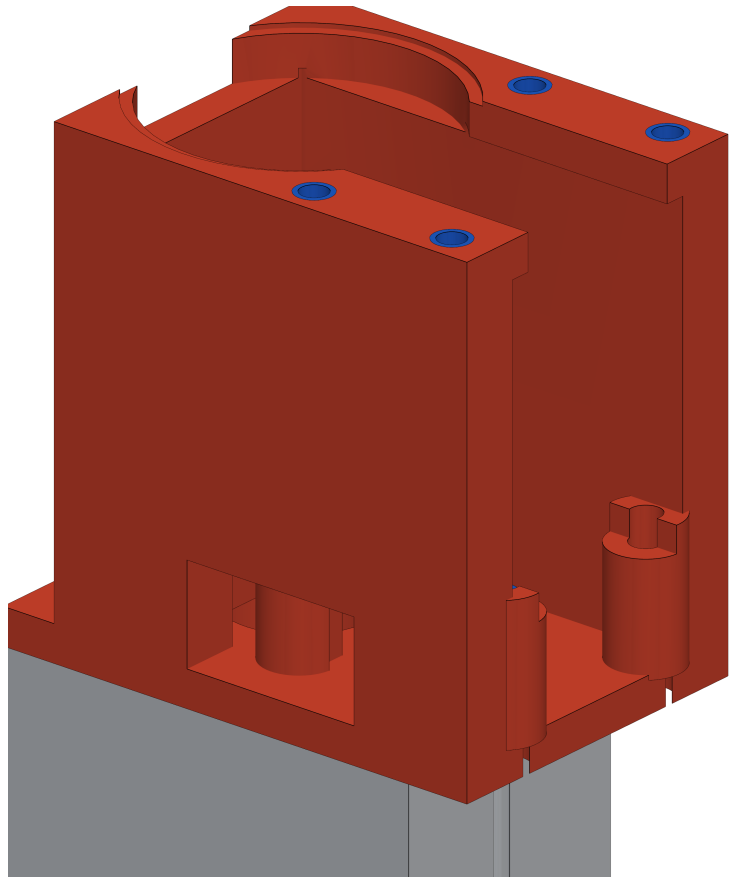
Step 14 of 24 (ON THE ARM)

J4 base onto the top of the forearm

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: j4_base
4 x M3 heat-set insert (set in preparation 2)
4 x M3 x 10 socket head
TOOLS: 2.5 mm hex key

DO

J4 base onto the forearm's end: onto the forearm's socket end, 4 x M3 x 10 (heads on the plate) into the forearm's end inserts. Lead the J3 -> J4 cable along the forearm to the J4 base.

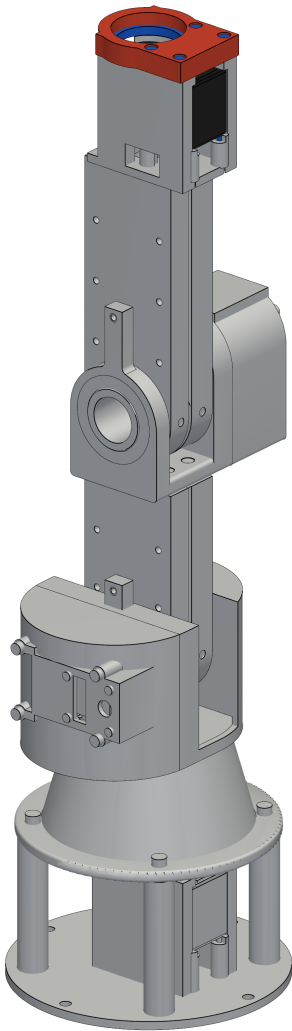
CHECK

The J4 base sits flat and square on the forearm.

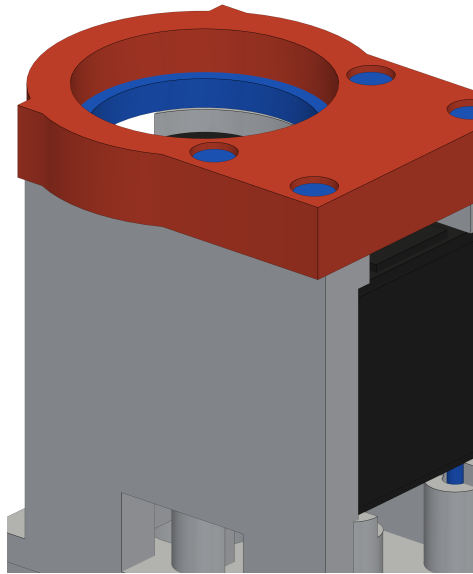
Step 15 of 24 (ON THE ARM)

J4 servo in (2 screws from under the plate), bearing, cap

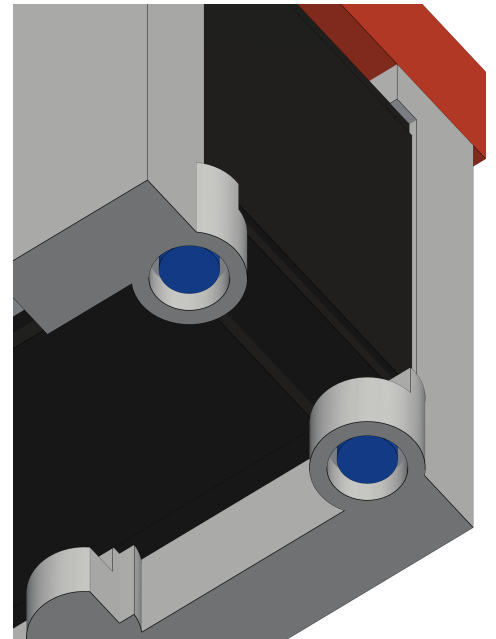
the arm so far



close up



from below (cut away)



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: j4_cap
1 x 6706 bearing (30x37x4) (pressed in: preparation 2)
4 x M3 x 12 socket head
1 x ST3215 servo
2 x self-tapping screw ~2.2 x 8 pan head (from the servo box)
TOOLS: 2.5 mm hex key, Phillips PH1 screwdriver, vise or clamp + retaining compound (e.g. Loctite 641)

DO

Plug the J3 -> J4 and J4 -> J5 cables into the J4 servo. Slide it into the J4 channel, horn up (along the forearm). Drive 2 self-tapping screws from UNDER the base plate's overhang into the servo's far back holes. Press the 6706 into the J4 cap (retaining compound), put the cap on the wall tops: 4 x M3 x 12, heads in the 3.8 mm counterbores.

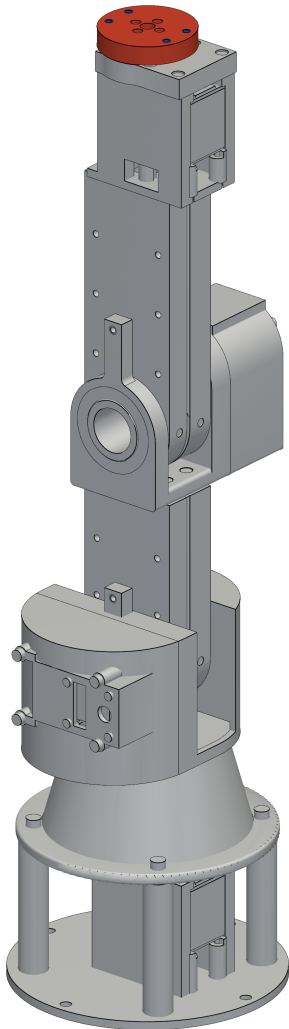
CHECK

The servo sits on its 4 posts; the cap's bearing is centred on the horn.

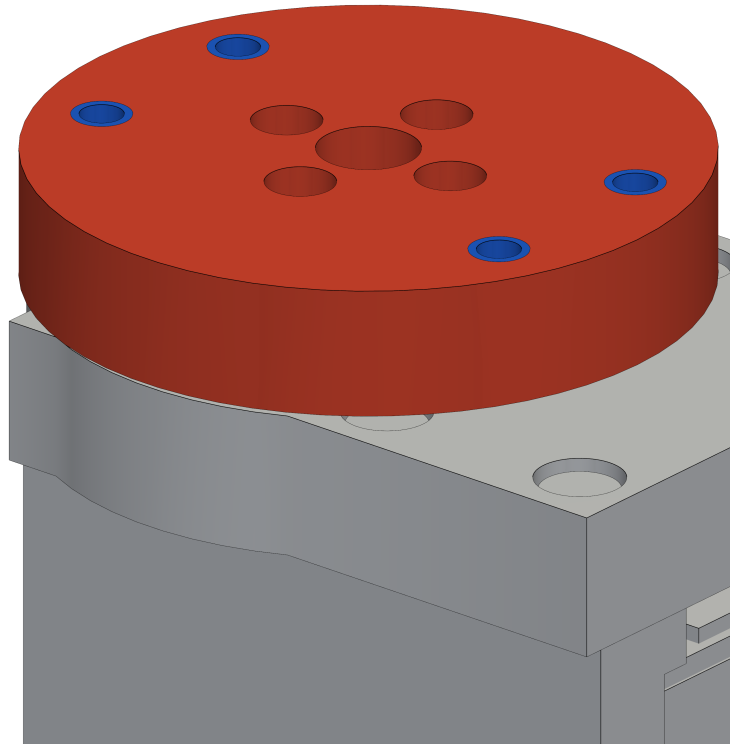
Step 16 of 24 (ON THE ARM)

J4 hub onto the J4 servo horn

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: j4_hub
4 x M2.5 x 6 socket head (horn)
4 x M3 heat-set insert (set in preparation 2)
TOOLS: 2.0 mm hex key

DO

Centre the J4 servo (2048); J4 hub onto the horn through the cap's bearing, 4 x M2.5 x 6.

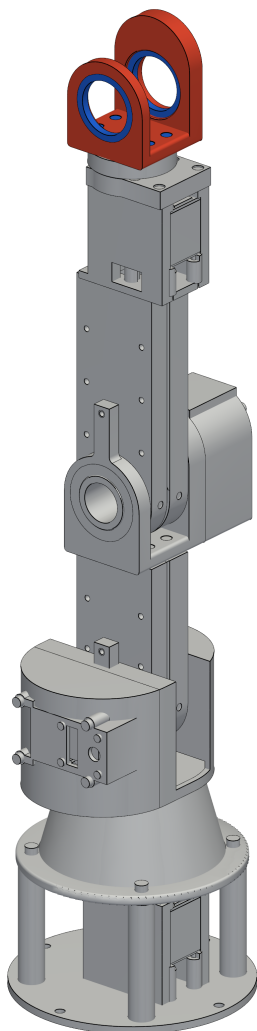
CHECK

The hub turns ± 90 deg without touching the cap (1.3 mm over the cap bolt heads).

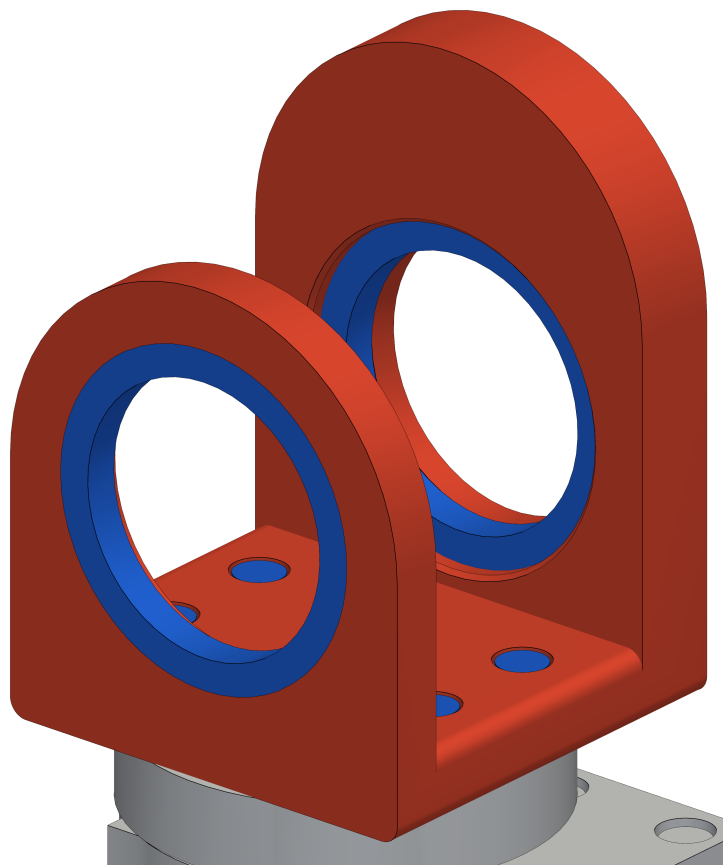
Step 17 of 24 (ON THE ARM)

J5 fork onto the J4 hub

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J5_p1

2 x 6706 bearing (30x37x4) (pressed in: preparation 2)

4 x M3 heat-set insert (set in preparation 2)

4 x M3 x 10 socket head

TOOLS: 2.5 mm hex key, vise or clamp + retaining compound (e.g. Loctite 641)

DO

J5 p1 has its two 6706 pressed in (preparation). Bolt it onto the J4 hub: 4 x M3 x 10 into the hub's inserts, heads sunk in the fork floor.

CHECK

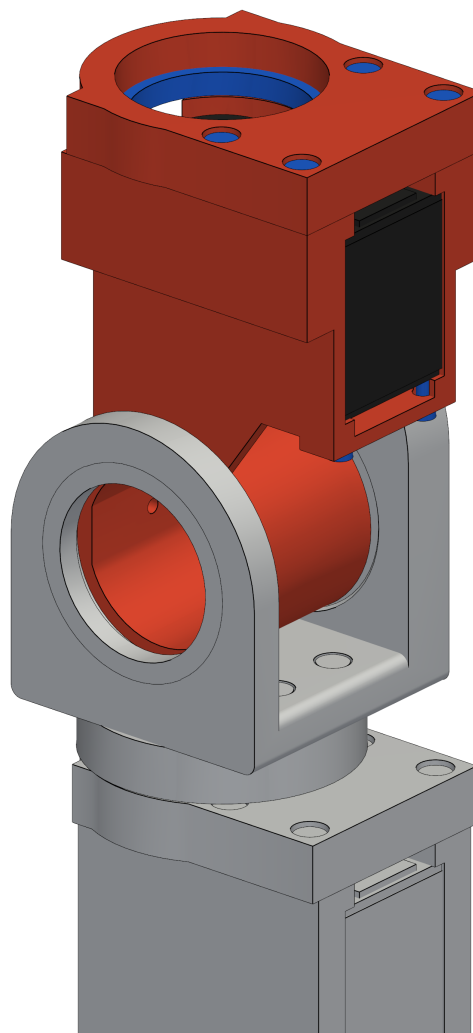
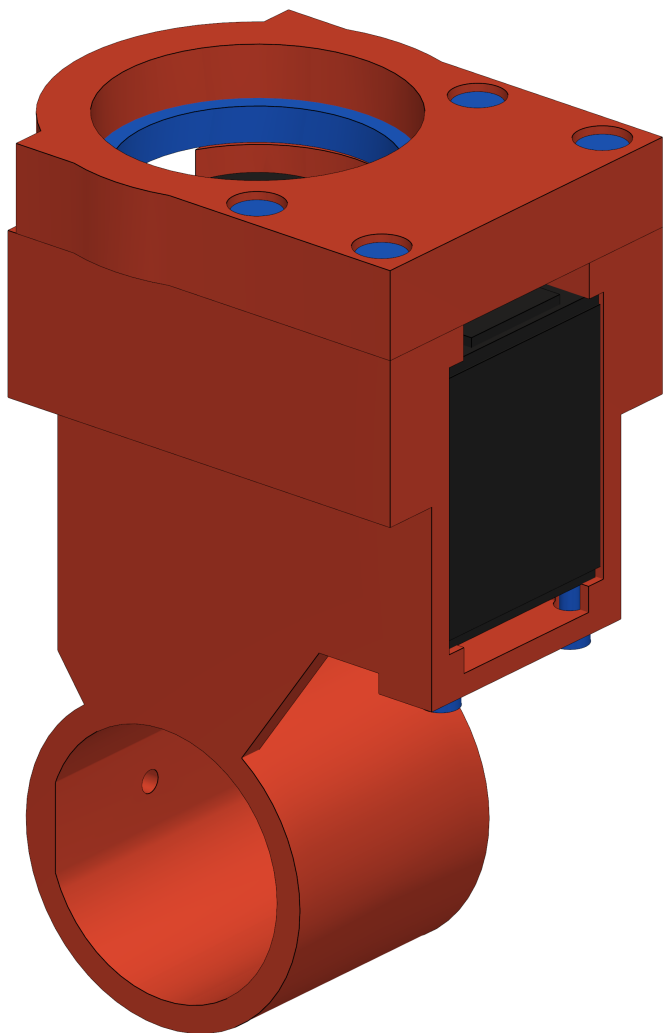
The fork sits flat on the hub; heads below the floor.

Step 18 of 24 (ON THE BENCH)

wrist blade -- J6 servo in, bearing, cap

on the bench

where it goes on the arm



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: j6_body, j6_cap

1 x 6706 bearing (30x37x4) (pressed in: preparation 2)

4 x M3 heat-set insert (set in preparation 2)

4 x M3 x 12 socket head

1 x ST3215 servo

2 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

TOOLS: 2.5 mm hex key, Phillips PH1 screwdriver, vise or clamp + retaining compound (e.g. Loctite 641)

DO

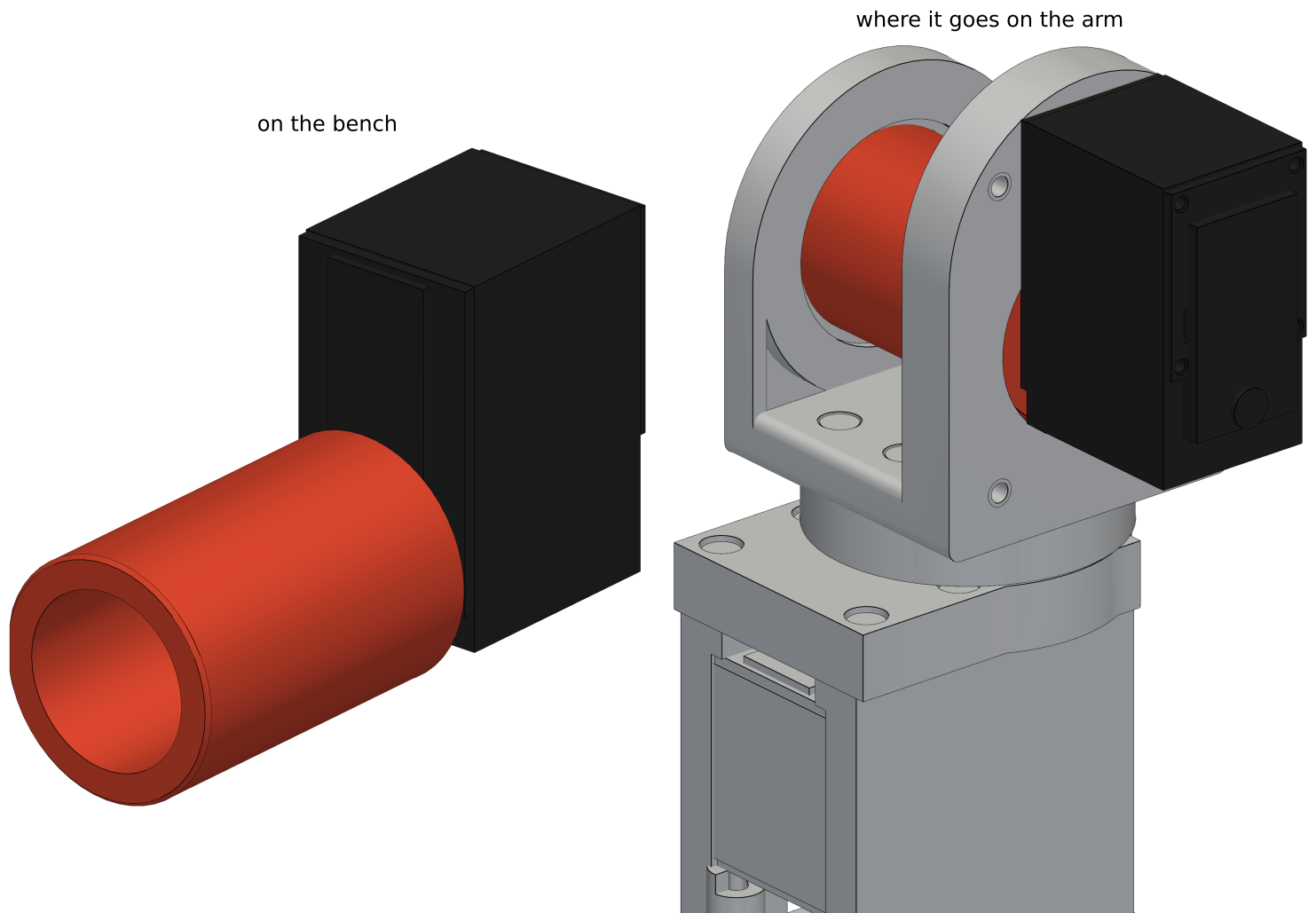
Wrist blade on the bench: plug the J5 -> J6 cable into the J6 servo; slide the servo into the blade's channel; 2 self-tapping screws through the far bridge into its back holes. Press the 6706 into the J6 cap, cap on: 4 x M3 x 12.

CHECK

The servo stands on its 2 rear pads and the bridge.

Step 19 of 24 (ON THE BENCH)

J5 servo onto its printed axle



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J5_shaft
4 x M2.5 x 6 socket head (horn)
1 x ST3215 servo
TOOLS: 2.0 mm hex key

DO

Centre the J5 servo (2048); J5 axle onto its horn, 4 x M2.5 x 6. Plug the J4 -> J5 and J5 -> J6 cables in.

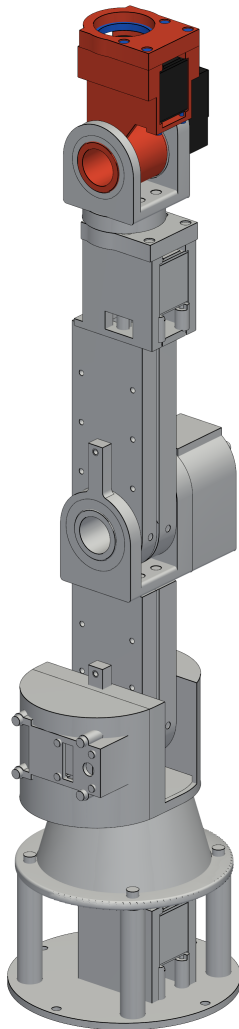
CHECK

The axle runs true.

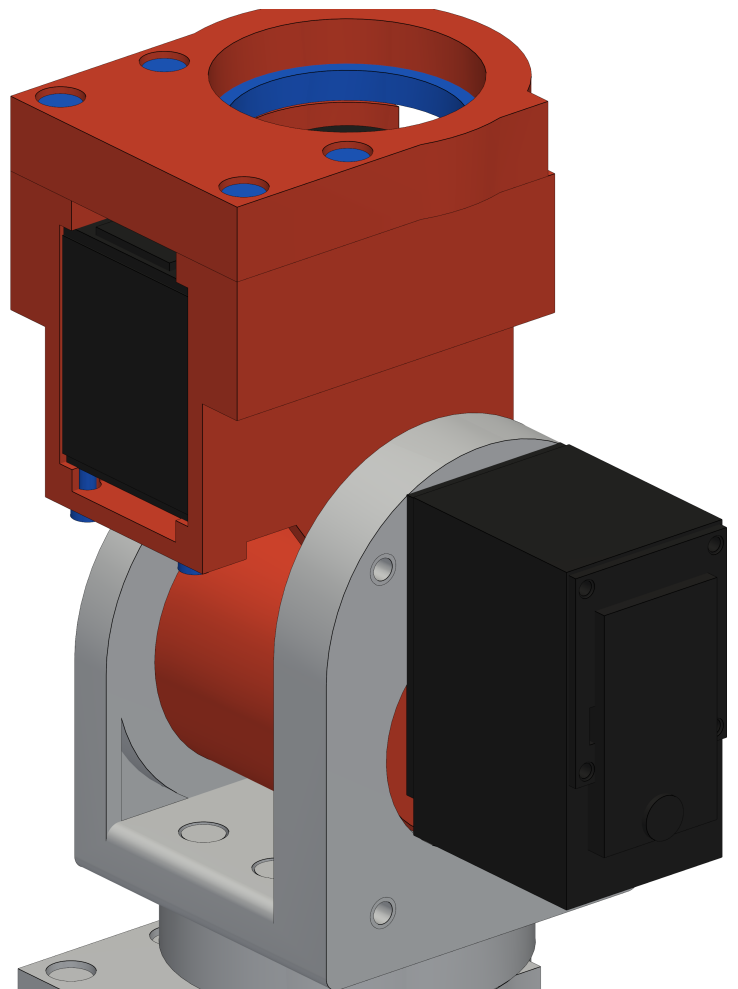
Step 20 of 24 (JOIN)

J5: blade into the fork with its spacers, axle in from the idle side

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J5_spacer_A, J5_spacer_B

DO

Blade into the J5 fork gap with a 3.0 mm spacer on each side between the blade and the inner races. Turn the axle's flat to the blade's D-bore and slide the axle in from the idle side; the J5 servo goes into its bay.

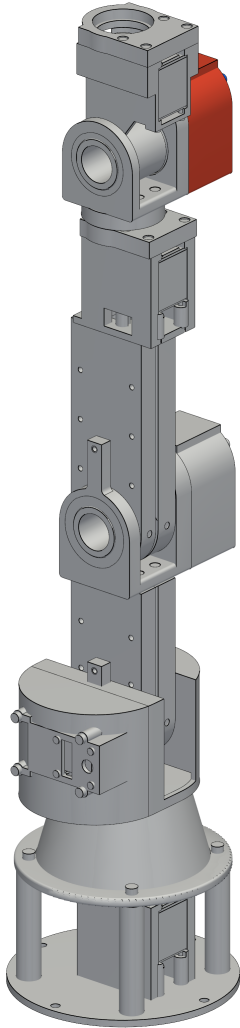
CHECK

The blade is clamped between the spacers without play; the axle is fully home.

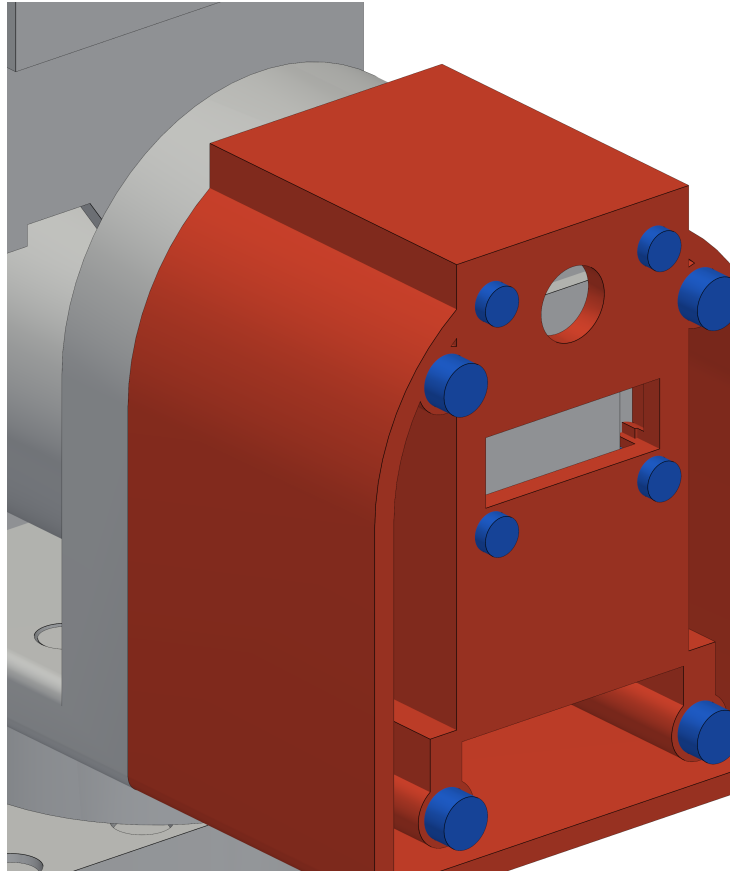
Step 21 of 24 (ON THE ARM)

J5: cover closed, servo screws, axle grub

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: J5_p2

1 x M3 x 4 set screw

4 x M3 x 45 socket head

4 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

TOOLS: 1.5 mm hex key, 2.5 mm hex key, Phillips PH1 screwdriver

DO

Cables out through the cover window; close the J5 cover: 4 x M3 x 45; 4 self-tapping servo screws; the M3 x 4 grub through the blade onto the axle's flat (1.5 mm key; it cuts its own thread in the 02.6 hole).

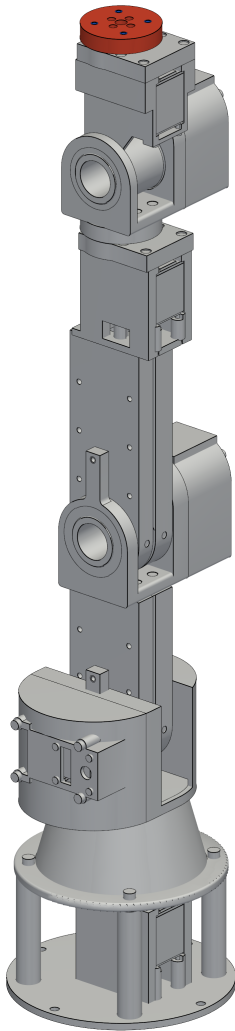
CHECK

The blade moves only with the J5 servo.

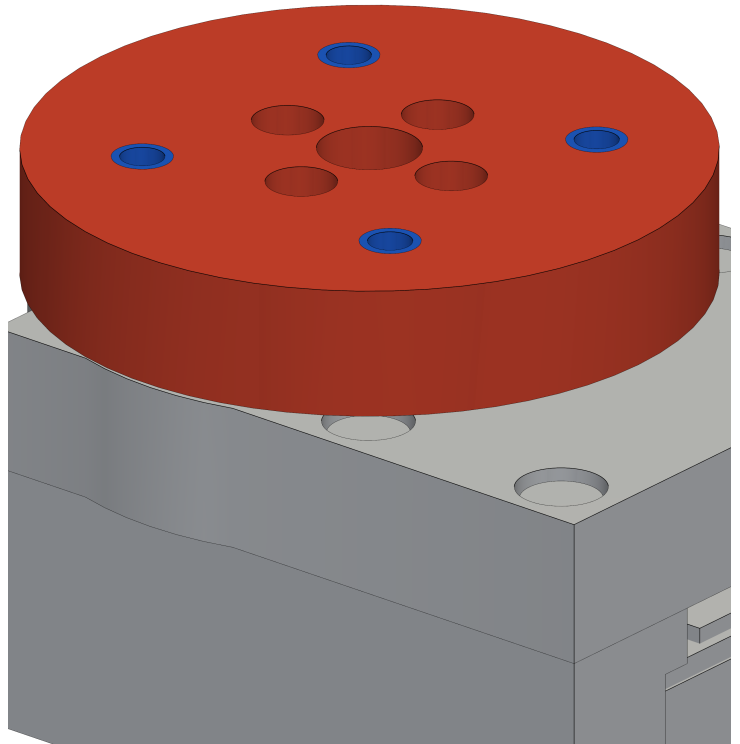
Step 22 of 24 (ON THE ARM)

J6: tool flange onto the J6 servo horn

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: j6_flange
4 x M2.5 x 6 socket head (horn)
4 x M3 heat-set insert (set in preparation 2)
TOOLS: 2.0 mm hex key

DO

Centre the J6 servo (2048); tool flange onto its horn through the cap's bearing, 4 x M2.5 x 6 from the tool side. The flange's 4 M3 inserts take your tool.

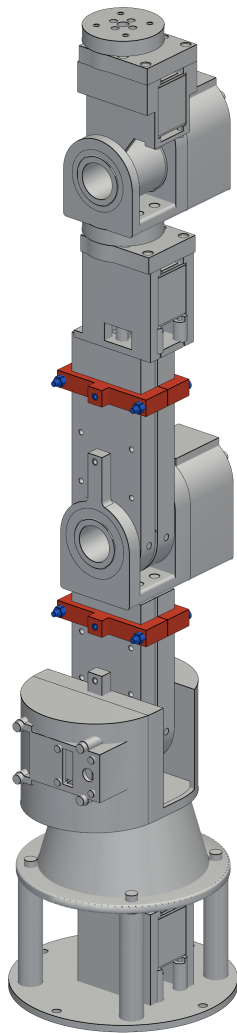
CHECK

The flange turns ± 90 deg without touching the cap.

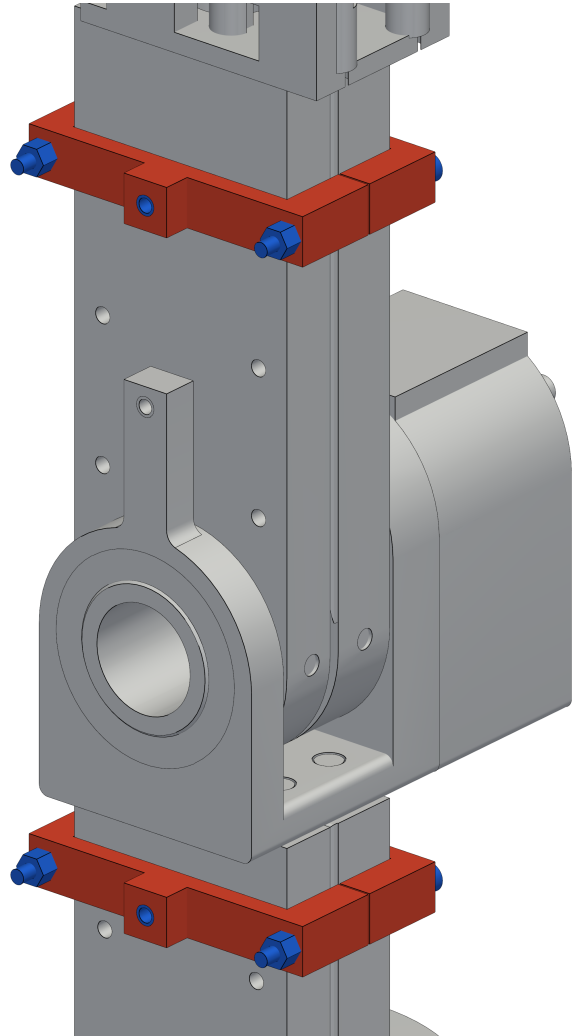
Step 23 of 24 (ON THE ARM)

Spring collars on the upper link and the forearm

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

printed: collar_fore_+y, collar_fore_-y, collar_upper_+y, collar_upper_-y
4 x M3 heat-set insert (set in preparation 2)
4 x M3 nyloc nut
4 x M3 x 45 socket head
TOOLS: 2.5 mm hex key, 5.5 mm spanner

DO

Spring collars round the upper link and the forearm: 2 halves each, 2 x M3 x 45 + nyloc nut per collar.

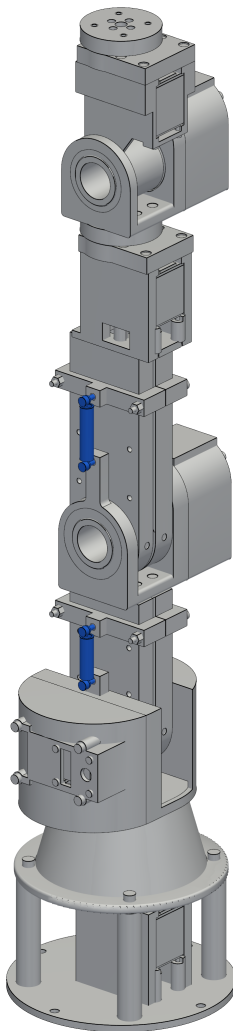
CHECK

The collars do not slide on the links.

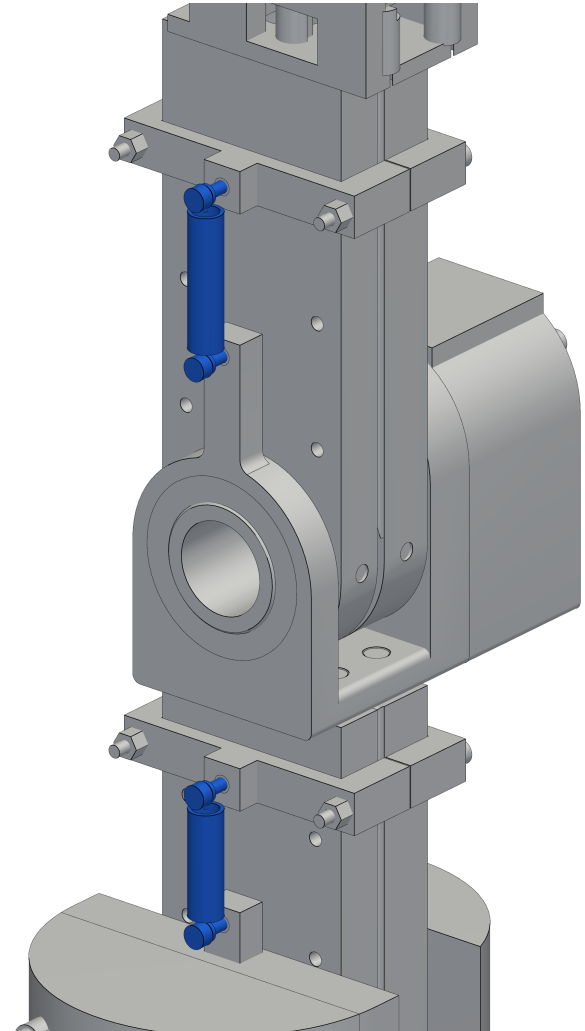
Step 24 of 24 (ON THE ARM)

Spring pins and the four gravity springs

the arm so far



close up



grey = already built red = new printed part black = new servo blue = new screws, bearings, inserts

YOU NEED

- 8 x M3 x 12 socket head
 - 2 x extension spring J2 (7.4 N @ 36.0 / 12.4 N @ 58.2 mm, inside hooks)
 - 2 x extension spring J3 (2.2 N @ 43.0 / 5.0 N @ 91.4 mm, inside hooks)
- TOOLS: 2.5 mm hex key

DO

Spring pins (M3 x 12 screws, 5.5 mm into their inserts, 6.5 mm stand out) into the turret, the J3 fork and the collars; hook the four springs (J2 pair, J3 pair). Now the arm holds itself up when unpowered.

CHECK

Every joint moves by hand over its whole range with the cables following in their loops.

Finished -- before the first power-up

joint	range (firmware limits)
J1_base_yaw	+ -90.0 deg
J2_shoulder	+ -54.0 deg
J3_elbow	+ -72.0 deg
J4_forearm_roll	+ -90.0 deg
J5_wrist_pitch	+ -46.5 deg
J6_tool_roll	+ -90.0 deg

- 1 Move every joint by hand through its whole range: nothing rubs, every cable follows in its loop and is tied down on both sides of its joint.
- 2 Check the six IDs answer on the bus (servo tool) -- one at a time is fine.
- 3 Set the ranges above as position limits in your software.
- 4 First moves: low speed, low torque limit, one joint at a time.
- 5 Straight up (home) is a singular pose for the wrist and elbow; start from a bent ready pose, e.g. J2 30, J3 -60, J5 30.
- 6 Watch the servo temperatures over the bus (J2 works hardest).

To take a servo out later: undo its steps (and the steps after them that cover it) in reverse order -- every servo's way out was checked free. The J4 servo's 2 screws are under the forearm spring collars: take the collars off first.